

Building Cloud-Native Data Architecture for Modern Applications

Digital applications rule our lives today. It's important to be familiar with the basic principles for building them.



Modern applications are based on digital technologies leveraging the cloud, data and AI. They are available 24/7, providing a rich, seamless and omni-channel user experience. Unlike legacy applications, which are custom built and are usually monolithic in nature, often accumulating a lot of technical debt over time, modern applications are reimagined to be cloud and AI native. They are scalable, resilient, and provide interfaces to connect with external systems. A good data framework is foundational to meet these requirements.

Cloud-native applications fall into various categories:

- A microservices-based web application is a set of small services, with each service doing a particular activity that can be scaled independently. Each of these services can be containerised and managed/deployed separately.
- An event-driven application can typically be triggered by an event and run as functions, pipelines or processes. The patterns are scheduled to meet the event triggering criteria.
- A data application ingests data in real-time/batch and stores it in a data lake. The data is processed further to make it ready for analytical

applications, which provide features like real-time dashboards.

- A chat application is a modern use case that can leverage the power of generative AI/agent AI tools and orchestrated workflows to answer user queries in natural language.
 - A mobile application typically runs on user handheld devices that typically have lower system configurations and provide anywhere access.
- Being cloud-native also enables:
- Continuous integration and continuous delivery, leading to faster time to market and fostering a culture of innovation and growth.
 - Infrastructure automation and DevOps, which are foundational for ensuring fit-for-use and fit-for-purpose on-demand infrastructure as well as collaboration between development and operational teams.
 - Ready-to-use identity and security components, which help to build applications that can handle users across the globe reliably with integrity.

Key principles for creating the data architecture

To meet the demands of modern cloud-native applications, organisations must have a strong data strategy.

The key principles are:

- Build a 'strong foundation'. Have a clear purpose and estimate the data needs of the applications.
- Adopt the 'data is gold' principle. Data from different sources and